environment, we would be fully able to generate still greater change. As our technology becomes more powerful, we will be able to work still more profound changes.

But just as (in parallel parking) it's easier to get out of a parking place than into one, it's easier to destroy a planetary environment than to move it into a narrowly prescribed range of temperatures, pressures, compositions, and so on. We already know of a multitude of desolate and uninhabitable worlds, and—with very narrow margins—only one green and clement one. This is a major conclusion from early in the era of spacecraft exploration of the Solar System. In altering the Earth, or any world with an atmosphere, we must be very careful about positive feedbacks, where we nudge an environment a little bit and it takes off on its own—a little cooling leading to runaway glaciation, as may have happened on Mars, or a little warming to a runaway greenhouse effect, as happened on Venus. It is not at all clear that our knowledge is sufficient to this purpose.

As far as I know, the first suggestion in the scientific literature about terraforming the planets was made in a 1961 article I wrote about Venus. I was pretty sure then that Venus had a surface temperature well above the normal boiling point of water, produced by a carbon dioxide/water vapor greenhouse effect. I imagined seeding its high clouds with genetically engineered microorganisms that would take CO_2, N_2, and H_2O out of the atmosphere and convert them into organic molecules. The more CO_2 removed, the smaller the greenhouse effect and the cooler the surface. The microbes would be carried down through the atmosphere toward the ground, where they would be fried, so water vapor would be returned to the atmosphere; but the carbon from the CO_2 would be converted irreversibly by the high temperatures into graphite or some other involatile form of carbon. Eventually, the temperatures would fall below the boiling point and the surface of Venus would become habitable, dotted with pools and lakes of warm water.

The idea was soon taken up by a number of science fiction authors in the continuing dance between science and science fiction